

Yining “Rima” Cao

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EDUCATION

University of California, San Diego

Ph.D. in Cognitive Science, Advisor: Haijun Xia

La Jolla, CA, US

2021 - present

University of Michigan, Ann Arbor

M.Sc. in Information Science, Advisor: Eytan Adar

Ann Arbor, MI, US

2019 - 2021

Tsinghua University

B.E. in Environmental Science · Minor in Computer Science, Advisor: Deyi Hou

Beijing, China

2015 - 2019

FULL PAPER & JOURNAL PUBLICATIONS

- 2026
(under review) Mingyi Li, Mengyi Chen, Sarah Luo, **Yining Cao**, Haijun Xia, Maitraye Das, Steven P. Dow, Jane L. E. 2026. Exploring Strategies for Displaying Computational Feedback in a Visual Design Tool. (title anonymized)
- 2026
(under review) Ruanqianqian (Lisa) Huang, Brian Hempel, **Yining Cao**, Haijun Xia, and Sorin Lerner. 2026. Always-Presentable Computational Notebooks. (title anonymized)
- 2025 [C.6] **Yining Cao**, Peiling Jiang, and Haijun Xia. 2025. Generative and Malleable User Interfaces with Evolving Task-Driven Data Schema. (CHI'25)
- 2025 [C.5] **Yining Cao**, Yiyi Huang, Anh Truong, Hijung Valentina Shin, and Haijun Xia. 2025. Compositional Structures as Substrates for Human-AI Co-creation Environment: A Design Approach and A Case Study. (CHI'25)
- 2025 [J.3] Doo Won Han, Hyesun Chung, **Yining Cao**, Feng Zhou, Lisa Molnar, Lionel P. Robert, Dawn M. Tilbury, X. Jessie Yang. 2025. A Systematic Review of Metrics Measuring Takeover Performance in Conditionally Automated Driving. International Journal of Human-Computer Interaction (2025): 1-30.
- 2025 [C.4] Bryan Min, Allen Chen, **Yining Cao**, and Haijun Xia. 2025. Malleable Overview-Detail Interfaces. (CHI'25)
- 2024 [C.3] **Yining Cao**, Ruabait Habib Kazi, Li-Yi Wei, Deepali Aneja, Haijun Xia. 2024. Elastica: Adaptive Live Augmented Presentations with Elastic Mappings Across Modalities. (CHI'24)
- 2023 [C.2] **Yining Cao**, Jane L E, Zhutian Chen, and Haijun Xia. 2023. DataParticles: Block-based and Language-oriented Authoring of Animated Unit Visualizations. (CHI'23)
 **Best Paper Award**
- 2022 [C.1] **Yining Cao**, Hariharan Subramonyam, and Eytan Adar. 2022. VideoSticker: A Tool for Active Viewing and Visual Note-taking from Videos. (IUI'22)
- 2021 [J.2] **Yining Cao***, Xiyue Jia*, David O'Connor, Jin Zhu, Daniel C W Tsang, Bin Zou, and Deyi Hou. 2021. Mapping soil pollution by using drone image recognition and machine learning at an arsenic-contaminated agricultural field. Environmental Pollution 270 (2021): 116281.

2020 [J.1] Liuwei Wang, Deyi Hou, **Yining Cao**, Yong Sik Ok, Filip M.G. Tack, Jörg Rinklebe, and David O'Connor. 2020. Remediation of mercury contaminated soil, water, and air: A review of emerging materials and innovative technologies. Environment International 134 (2020): 105281.

WORKSHOP & EXTENDED ABSTRACT

2026 [EA.2] Siân Lindley, Jack Williams, **Yining Cao**, Haijun Xia, Elizabeth F Churchill, Abigail Sellen, Jeffrey Nichols, David R Karger. 2026. What does Generative UI mean for HCI Practice? (CHI EA'26)

2021 [EA.1] **Yining Cao**, Feng Zhou, Elizabeth M Pulver, Lisa J Molnar, Lionel P Robert, Dawn M Tilbury, and Xi Jessie Yang. 2021. Towards standardized metrics for measuring takeover performance in conditionally automated driving: A systematic review. In Proceedings of the Human Factors and Ergonomics Society Annual Meeting. (HFES'21)

PROFESSIONAL EXPERIENCE

Microsoft Research , Research Scientist Intern Mentor: Nathalie Riche	2025 Summer
Adobe Research , Research Scientist Intern Mentor: Anh Truong	2023 Summer
Adobe Research , Research Scientist Intern Mentor: Rubaiat Habib Kazi	2022 Summer
Interaction & Collaboration Research Lab, University of Michigan , Research Assistant Mentor: Xi Jessie Yang	2020 - 2021
Fitly Inc. , Machine Learning Engineer Intern Mentor: Honglak Lee	2020 Summer

SELECTED AWARDS

2021 - 2025	Kroner Fellowship, UC San Diego
2023, 2024	Special Recognitions for Outstanding Reviews for UIST
2023	ACM CHI Best Paper Award
2019	Honored Graduate Award, Tsinghua University
2018	Zheng Geru Scholarship, Tsinghua University
2017	The Veolia Special Scholarship, Tsinghua University
2017	First Prize, National Mathematical Modeling Contest
2016	Outstanding Student Leader, Tsinghua University

SERVICE AND LEADERSHIP

2025	Publication Co-Chair. ACM Conference on User Interface Software and Technology (UIST)
2024 - 2025	Reviewer. ACM Conference on User Interface Software and Technology (UIST)
2023 - 2025	Reviewer. ACM Conference on Human Factors in Computing Systems (CHI)
2023 - 2024	Reviewer. IEEE Advances in Visualization and Visual Analytics (VIS)

2022	Student Volunteer. ACM Conference on Intelligent User Interfaces (UIST)
2022	Student Volunteer. ACM Conference on Intelligent User Interfaces (IUI)
2020 - 2021	Reviewer. International Conference on Information Systems (ICIS)
2017 - 2018	President of Rural International Student Exchange Association (RISE)

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant

TA, DSGN 118: Design for Creativity and Productivity	Fall 2025, Winter 2024, Fall 2024
TA, COGS 120: Introduction to Human-Computer Interaction	Fall 2023, Fall 2022, Winter 2022
TA, DSGN 100: Prototyping	Winter 2023

Student Mentored

Holly Zhu	Computer Science Undergraduate, UCSD Now Master's at CMU
Keshav Tiwari	Cognitive Science Undergraduate, UCSD
Yiyi Huang	Computer Science Undergraduate, UCSD Now SWE at Google